AVIRAL MALIK

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SUMMARY

Computer Science undergraduate at VIT Vellore with a strong foundation in data structures, algorithms, and object-oriented programming. Four internships building production backend services, REST APIs, and cloud deployments with Python, C++, FastAPI, Docker, and AWS, including agentic AI workflows with LLM agents and RAG pipelines.

EDUCATION

B.Tech in Computer Science Engineering (AI & ML) | Vellore Institute of Technology, Vellore **2024 – 2028**
CGPA: 9.73/10 | Ranked 3rd in the Department of CSE-AIML | Merit Scholarship recipient

Grade XII (CBSE) | National Public School, Bengaluru | 93.6% **2023 – 2024**

WORK EXPERIENCE

AI Engineer Intern | Xibotix Private Limited, Tamil Nadu **Jul 2026 – Aug 2026**
- Built edge AI and LLM applications, optimizing and deploying deep learning models on embedded platforms with C++.
- Contributed to machine learning research, model optimization, and production AI pipelines using Python and PyTorch.

AI Agent Development Intern | SkillWiz, Remote **May 2026 – Jul 2026**
- Architected a Generative AI grading engine with FastAPI, Gemini, and Whisper to score submissions and speech confidence.
- Engineered bidirectional REST APIs linking the FastAPI engine to a NestJS backend to deploy autonomous AI agents.
- Enforced LLM guardrails with Pydantic validation; containerized the service with Docker and deployed it on AWS EC2.

AI & ML Intern | Kineton Technologies, Tamil Nadu **May 2025 – Aug 2025**
- Engineered C-based image processing and compression algorithms optimized for resource-constrained hardware.
- Designed and simulated SystemVerilog hardware accelerators in ModelSim to speed up ML inference, including decision trees.

AI & ML Intern | Voxvertex, Tamil Nadu **Mar 2025 – Apr 2025**
- Deployed an LLM-powered website chatbot (Flask, Hugging Face Transformers) for company information and site navigation.
- Created an NLP semantic search tool to match guest lecturers to events; prototyped text-to-image visuals with diffusion models.

PROJECTS

Multi-Agent Customer Support System | Agentic AI Workflow **Oct 2026**
Python, LangGraph, LangChain, Llama 3 (Ollama), FAISS, Sentence-Transformers, SQLite
- Built a LangGraph multi-agent workflow routing queries to 5 specialized agents with RAG over a FAISS vector database.
- Achieved 92% intent-routing accuracy on a 50-query labeled eval set; added SQLite memory and human-in-the-loop approval.

Reflexive RAG-RTL | AI-Enhanced RISC-V Hardware Generation **Apr 2026**
Python, Qwen2.5-Coder (Ollama), ChromaDB, Sentence-Transformers, Verilog, Verilator
- Designed a Reflexive RAG pipeline on a ChromaDB vector database that eliminated LLM hallucinations, cutting lint errors 22 to 2.
- Generated a complete RISC-V core passing 38/38 ISA tests using a trust hierarchy that prioritizes verified module interfaces.

LabFlow | Document-to-Workflow Procurement Automation **Sep 2026**
React, TypeScript, AWS Lambda, Step Functions, DynamoDB, Cognito, Textract, PaddleOCR
- Developed a serverless AWS app turning procurement forms into role-based approval workflows with confidence-aware OCR.
- Implemented quotation comparison, policy validation, role-based access control, and an append-only audit trail.

Conflict-Aware Guest Selection | Academic Research **Mar 2026**
Python, Branch and Bound, Lagrangian Relaxation, Dynamic Programming
- Solved an NP-hard Quadratic Knapsack Problem exactly, exploring under 1.5% of the search space for up to a 70x speedup.
- Recovered the provably optimal solution in 40/40 configurations where Greedy and DP baselines produced negative scores.

TECHNICAL SKILLS

Languages: Python, C/C++, Java, TypeScript, SQL
ML & Deep Learning: PyTorch, TensorFlow, Hugging Face Transformers, scikit-learn, XGBoost, OpenCV, Pandas, NumPy
LLMs & GenAI: RAG, LLM Agents, LangChain, Prompt Engineering, Fine-Tuning, Vector Databases (ChromaDB), Embeddings, Ollama
Tools & Cloud: FastAPI, Flask, REST APIs, Docker, AWS, MongoDB, Git, Streamlit, SystemVerilog, Verilator
Concepts: Machine Learning, Deep Learning, NLP, Computer Vision, Generative AI, Edge AI, Data Structures & Algorithms, OOP

ACHIEVEMENTS & CERTIFICATIONS

- Ranked 3rd in the CSE-AIML department at VIT (Merit Scholarship); CBSE School Topper in Grade XII (3rd overall, 1st in Chemistry); 3rd place in three national-level karate championships (2017–2018).
- Certifications: Deep Learning Specialization and Machine Learning Specialization (DeepLearning.AI, Stanford); Generative AI Engineering with LLMs, Building Agents with Agentic AI, Python for Data Science, AI & Development (IBM).